

Replay Channel with JUNA and Other Baselines

Sensitivity and Tuning Practical Advice

July 10, 2026

Abstract

This paper gives practical tuning advice for replaying measured underwater acoustic channels with JUNA+FEC OFDM and two baselines, OFDM+FEC and Partial-FFT+FEC. The basic question is simple: if a modem setting works on one measured channel, which parts of that setting can be reused on another channel, and which parts must be retuned? We test twelve full-duration replay captures: **SG-1/2/3**, **NA-1/2/3**, and **HW-1-6**. For each channel we run a small coordinate sweep at 20 dB, plus one-factor-at-a-time (OAT) sensitivity sweeps over 10/20/30 dB. In an OAT sweep, one modem parameter is changed while the others are held fixed, so the result is a local tuning guide, not a global proof of parameter importance.

We compare three reference points because they answer different practical questions. The *cold default* is the same untuned starting configuration on every channel; it tells us what happens before channel-specific tuning. The *fixed winner* is one reusable cross-channel recipe, frozen and run unchanged everywhere; it tells us how far a shared setting can transfer. The *own optimum* is each channel’s best setting from the finite 20 dB search; it is a per-channel benchmark, not a reusable recipe.

The main result is that the modem has many knobs, but they do not all matter at the same time. A compact family transfers well: QPSK, FEC rate 0.5, and outer pilot ratio 1/5 are good defaults across most channels. The setting that most often must be retuned first is the FFT size N . From the cold default, changing N is the largest way to move packet success on the harder SG and NA captures. Once N has been chosen well, however, its importance collapses, and the dangerous knobs become the ones that can break an already-working link, especially high FEC rate and 8PSK. The practical advice is therefore two-stage: first search N so packets start decoding; then keep FEC and modulation conservative enough that the receiver stays there.

1 What Was Run

Every row in this paper is a full-capture replay result. That means `segments=0`: the receiver decodes every frame available in the measured recording, not a short hand-picked excerpt. A packet is counted as successful only if its payload CRC passes. The two main metrics are therefore:

$$\text{PSR} = \frac{\text{accepted packets}}{\text{decoded packets}}, \quad \text{BER} = \text{post-FEC message bit-error rate.}$$

Configurations are ranked PSR-first, with BER as the tie-breaker.

The captures come from three measured deployment sites, which we abbreviate **SG**, **NA**, and **HW**:

- **SG — Singapore harbour**: shallow water, 0.1–0.4 km, ≈ 47.8 s, 25 kHz carrier, 9.6 kHz baseband. Captures SG-1/2/3.

- **NA — North Atlantic:** mobile deep-water, 3–7 km, ≈ 52 s, 13 kHz carrier, 4.9 kHz baseband. Captures NA-1/2/3.
- **HW — Hawaii:** moored deep-water, 3 km, 13 kHz carrier, 6.25 kHz baseband. Six captures HW-1...6 with a wide range of difficulty.

The three sites are physically distinct, which is what ultimately sets the channel-difficulty range seen throughout this paper. Figure 1 shows a representative measured time-varying channel impulse response (CIR) from one capture per site.

The number of decoded packets also depends on N : roughly 630 packets at $N = 512$, 360 at $N = 1024$, and 190 at $N = 2048$. This is why the tables focus on PSR, not just raw packet count.

We use four increasingly specific checks. A best-config sweep runs a 32-configuration coordinate search at 20 dB for each channel to find a good operating point. A cold-default OAT sweep starts from the same default config on every channel, moves one parameter at a time, and records how much PSR changes. Interaction grids on SG-1 and NA-3 sweep N together with CP, FEC, and pilot spacing, and an optimum-centered OAT sweep then repeats the one-at-a-time test around each channel’s own best config.

2 First Answer: What Config Works Best?

Table 1 reports the best JUNA configuration found for each channel at 20 dB under the finite search used in this study. The search is a staged brute-force coordinate search rather than an exhaustive Cartesian sweep. It starts from the common reference setting

$$\begin{aligned} N &= 2048, & \text{CP} &= 64, & \text{symbols/frame} &= 10, \\ \text{outer pilot} &= 1/4, & \text{inner pilot} &= 1/4, & \text{FEC} &= 0.25, \\ d_c &= 8, & \text{modulation} &= \text{QPSK}. \end{aligned}$$

For each channel, 32 candidate settings are evaluated:

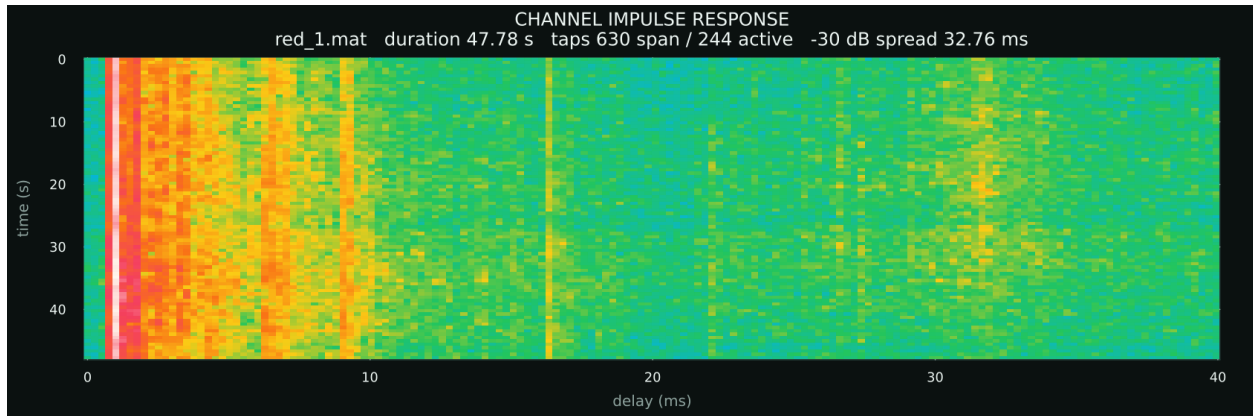
$$\begin{aligned} \text{FEC} &\in \{0.25, 0.5, 0.75\}, \\ (\text{outer pilot}, \text{inner pilot}) &\in \{1/10, 1/7, 1/5\} \times \{1/10, 1/7, 1/5\}, \\ d_c &\in \{6, 8, 10\}, \\ (N, \text{CP}) &\in \{512, 1024, 2048\} \times \{0, 16, 32, 64, 128\}, \\ \text{modulation} &\in \{\text{QPSK}, 8\text{PSK}\}. \end{aligned}$$

After each stage, the PSR-best setting, with BER as the tie-breaker, is carried forward to the next stage. Thus Table 1 should be read as the best setting found within this finite search budget, not as a proof of global optimality.

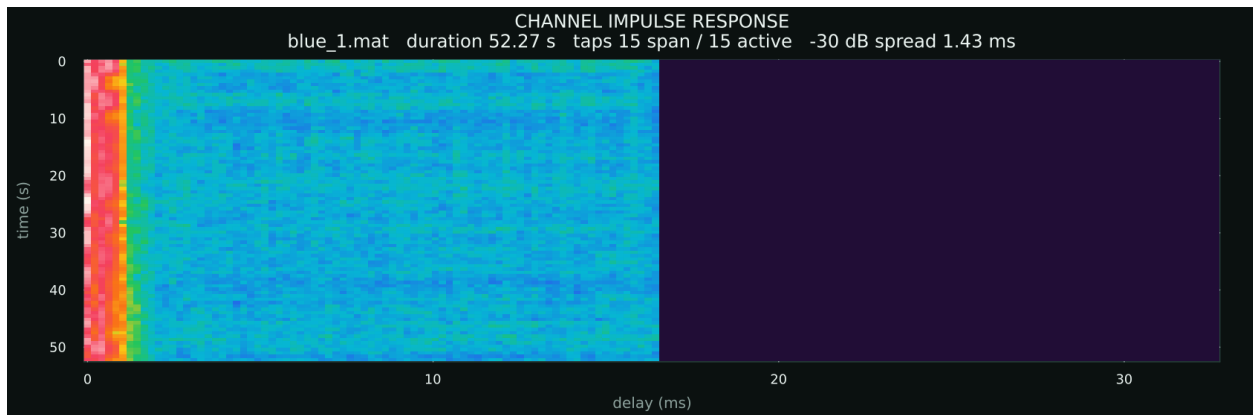
The resulting winners show a consistent but not universal pattern. QPSK, FEC rate 0.5, and moderately dense pilots appear repeatedly, while N , CP, and d_c remain channel dependent. Four captures reach perfect packet success under the selected configuration (SG-1, NA-1, HW-1, and HW-4), but several others remain below perfect PSR even after tuning. The configuration search therefore improves the operating point, but it does not remove the underlying differences in channel difficulty.

3 Transfer Without Retuning: One Config on Every Channel

The per-channel winners in Table 1 assume that we have enough time to tune every new capture. In practice, that may not be true. A more operational question is: if previous channel experience



(a) **SG-1** — Singapore harbour: shallow-water front arrival with a low-level tail.



(b) **NA-1** — North Atlantic: compact response despite the mobile deployment.



(c) **HW-1** — Hawaii: compact main cluster (later HW captures have longer delay spreads).

Figure 1: Representative measured time-varying CIRs, one capture per site (full text width for legibility). Delay is horizontal, experiment time is vertical, and colour is normalised tap strength.

Table 1: Best full-capture configuration at 20 dB (PSR-first, BER tie-break). *Pilot* is the total pilot ratio (outer + inner pilot fraction), directly comparable to the single pilot grid OFDM and Partial-FFT use; d_c is the derived LDPC check degree; *rate* is the effective delivered payload throughput (accepted packets $\times$ payload bits $\div$ capture duration).

channel	N	CP	outer pilot frac.	FEC	d_{cfg}	mod	payload-exact	PSR	BER	acc./pkts.	rate (bit/s)
SG-1	2048	64	0.10	0.5	8	QPSK	1.000		0	10/10	5540
SG-2	512	0	0.20	0.5	8	QPSK	1.000		0	10/10	3078
SG-3	2048	64	0.25	0.5	8	QPSK	1.000		0	1/1	1308
NA-1	512	0	0.20	0.5	6	QPSK	1.000		0	10/10	2751
NA-2	512	0	0.14	0.5	8	QPSK	1.000		0	1/1	793
NA-3	512	0	0.14	0.5	8	QPSK	1.000		0	1/1	428
HW-1	2048	64	0.25	0.25	8	QPSK	1.000		0	10/10	1565
HW-2	2048	64	0.25	0.25	8	QPSK	1.000		0	10/10	3985
HW-3	2048	64	0.25	0.25	8	QPSK	1.000		0	10/10	3303
HW-4	2048	64	0.25	0.25	8	QPSK	1.000		0	10/10	1546
HW-5	512	0	0.20	0.5	6	QPSK	1.000		0	10/10	2254
HW-6	1024	0	0.20	0.5	6	QPSK	1.000		0	10/10	2024

has already given us a known-good configuration, what penalty do we pay for reusing it on a new channel without retuning? To answer that question, Figure 2 compares three reference points:

Cold default: an untuned, common starting config applied to every channel. “Cold” means before the channel-specific search has done any work. It asks: *what happens if we use a reasonable but generic modem setting?*

Fixed winner: one reusable config chosen as the best cross-channel default, then frozen and run unchanged on every channel. It asks: *how much can a single shared recipe buy us?*

Own optimum: each channel’s best config found by the 20 dB search in Table 1. It asks: *how well can this channel do if we allow channel-specific tuning?* This is a search-budget upper benchmark, not a transferable setting.

The fixed winner is:

$$N = 1024, \quad \text{CP} = 32, \quad \text{FEC} = 0.5, \quad \text{outer pilot} = 1/5, \quad d_c = 10, \quad \text{QPSK}.$$

For reference, we also run the cold default ($N = 2048$, $\text{CP} = 64$, $\text{FEC} = 0.25$, QPSK) with identical 10-symbol framing on every channel.

In Figure 2, the blue bar is therefore the key transfer test: it shows how far a single reusable config can go before channel-specific tuning is allowed.

The fixed winner beats the cold default on every channel and comes within 0.10 PSR of the channel optimum on eight of twelve channels. That is good news: one compact default transfers surprisingly well. But the misses are important. NA-2, NA-3, HW-6, and SG-2 need a different N and/or CP. In particular, NA-3 needs $N = 512$; forcing $N = 1024$ drops it to zero packet success.

So the fixed config is a strong starting point, not a universal answer.

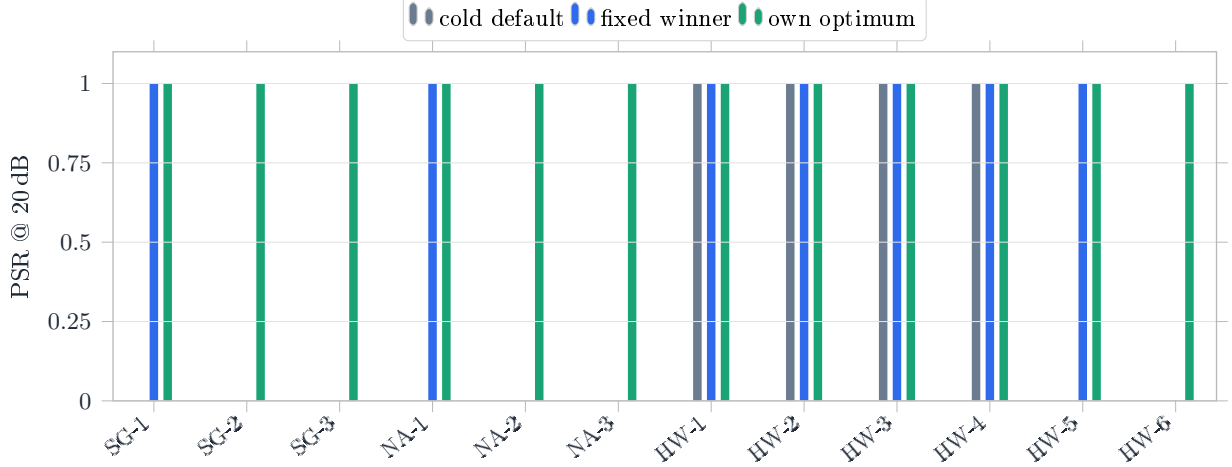


Figure 2: Transfer test at 20 dB. Grey: cold default, the untuned common starting point. Blue: one fixed recipe run unchanged on every channel. Green: per-channel search result from Table 1. Blue beats grey everywhere and reaches green on 8/12 channels; the misses need a different N /CP.

4 Does JUNA Help More Than Partial FFT?

Now keep the waveform fixed and change only the receiver. All three methods decode the same packets at the same N , CP, bandwidth, modulation, and pilot pattern:

$$\text{OFDM+FEC}, \quad \text{Partial FFT+FEC}, \quad \text{JUNA+FEC}.$$

Figure 3 shows the result at the fixed winner config.

There are three regimes. In the **hard-but-decodable** regime, JUNA helps most. On SG-1, PSR improves

$$0.26 \text{ (OFDM)} \rightarrow 0.72 \text{ (Partial FFT)} \rightarrow 1.00 \text{ (JUNA)}.$$

On NA-1, it improves $0.50 \rightarrow 0.86 \rightarrow 0.95$. This is the regime where decoder feedback is useful equalizer training information.

In **ceiling** captures, such as HW-1/2/4, the baselines are already near perfect, so JUNA has little room to improve. In a **broken-config** case such as NA-3 with forced $N = 1024$, all three fail. JUNA cannot rescue a geometry that is wrong for the channel.

Across all twelve channels, mean PSR rises from 0.49 for OFDM+FEC to 0.56 for Partial FFT+FEC and 0.66 for JUNA+FEC. Partial FFT is usually better than plain OFDM, but not always. The robust statement is that JUNA is at least as good as the better baseline on this fixed-config test.

Figure 4 adds the SNR dimension for two representative channels. The rationale is to separate a single 20 dB snapshot from a trend: with the waveform geometry fixed, do the receiver rankings persist as the additive-noise level changes? The answer is yes for the two channels shown. On SG-1, JUNA is already near the top of the plot at 10 dB (0.94 PSR), while Partial FFT reaches 0.51 and OFDM reaches 0.18; at 20–30 dB JUNA saturates at 1.00, while the baselines remain below it. On NA-1, the same ordering holds but the gap is smaller: $0.48 \rightarrow 0.69 \rightarrow 0.86$ at 10 dB and $0.52 \rightarrow 0.86 \rightarrow 1.00$ at 30 dB for OFDM, Partial FFT, and JUNA respectively. The figure is therefore a receiver comparison under controlled waveform and replay conditions, not a full channel-capacity or noise-model claim.

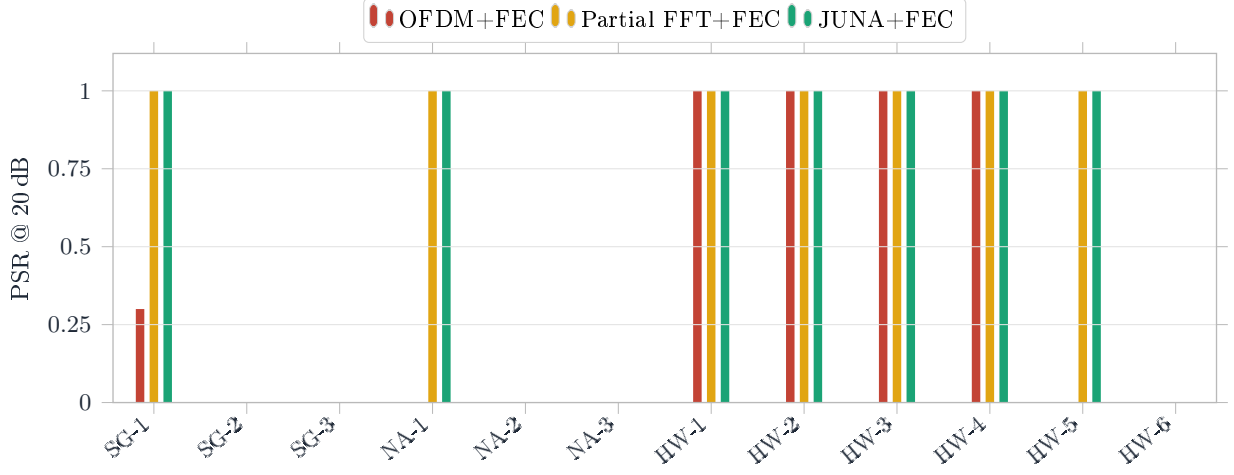


Figure 3: Three receivers on the same waveform (winner config, 20 dB). JUNA is highest on hard-but-decodable channels. All three converge on ceiling captures and fail together when the geometry is wrong for the channel.

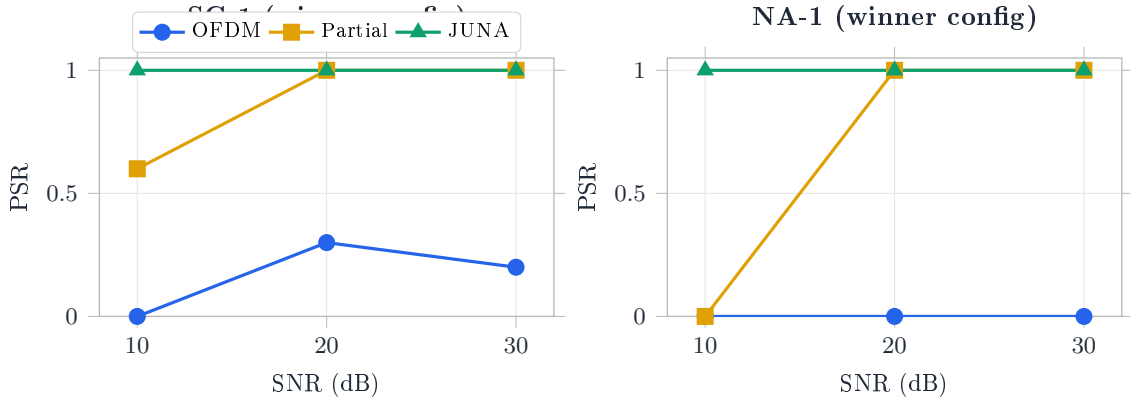


Figure 4: PSR vs SNR at the winner config for two representative channels. The figure compares receiver ranking under the same waveform and replay condition; it is not intended as a full channel-capacity explanation.

5 Which Knob Matters First?

The OAT sweep asks a simple question: starting from the same cold default, if we move only one knob, how much does packet success change? Table 2 reports that PSR swing.

The main reading is that N is the largest first determinant. It controls the OFDM time-frequency geometry, so when N is mismatched the receiver may not reach the packet-success floor at all. After N is set well enough for packets to decode, the limiting knob is no longer universal: some channels become pilot-limited, some become FEC-limited, and some become modulation-limited. This is plausible physically. A relatively static channel can tolerate a different pilot/FEC tradeoff than a moving or rapidly varying channel, where pilot density and coding margin can decide whether an otherwise working geometry stays locked.

Table 2: Cold-default OAT mean PSR swing by parameter (avg over 10/20/30 dB). Bold marks the rank-1 lever per channel. N is the largest first determinant on most non-ceiling captures; after the geometry is working, the bottleneck can shift to pilots, FEC, or modulation depending on channel behaviour.

	SG-1
N	1.0
d_{cfg}	.00
outer pilot	1.0
FEC	.00
mod	.00
CP	.00

6 Does “Tune N First” Always Hold?

OAT results are local. They tell us what matters near the starting point, not what matters everywhere. The cold default uses $N = 2048$, which is bad for several hard channels, and FEC 0.25, which is conservative. So it makes sense that changing N helps most from that starting point.

The next question is: after N is already good, does N still dominate? We test that two ways.

We repeat the OAT sweep, but now the starting point is each channel’s own best config from Table 1. The dominant lever changes once the modem is already near a good setting. From the cold default, N is rank-1 on all six SG/NA channels. Around the optimum, FEC becomes rank-1 on four channels, modulation on NA-1, and N remains rank-1 only on NA-2. The reason is that once N is well chosen, moving it does less: on SG-1, N ’s PSR swing drops from 0.68 to 0.21, and on NA-1 it drops from 0.55 to 0.09. Meanwhile FEC becomes the dangerous knob: on SG-1, its swing rises from 0.32 to 1.00. At a good operating point, the important knobs are the ones that can break a working link, especially too high a code rate or too aggressive a modulation.

Figure 5 shows interaction grids for a clean channel (SG-1) and a hard channel (NA-3). The grid answers a practical concern: maybe N only looks important because CP or pilot spacing was wrong. The result is reassuring but nuanced.

On SG-1, $N = 1024$ is best across the CP range. On NA-3, $N = 512$ is best across the CP range. So N really can be tested first. But the best N is not the same for every channel. A rule like “always use $N = 1024$ ” works on SG and fails on hard NA.

The harder channel also shows coupling. On NA-3, $N = 512$ only works when FEC and pilot density are conservative enough. FEC 0.75 is fatal at every N . This matches the previous section: first choose N , then guard the FEC and modulation floors.

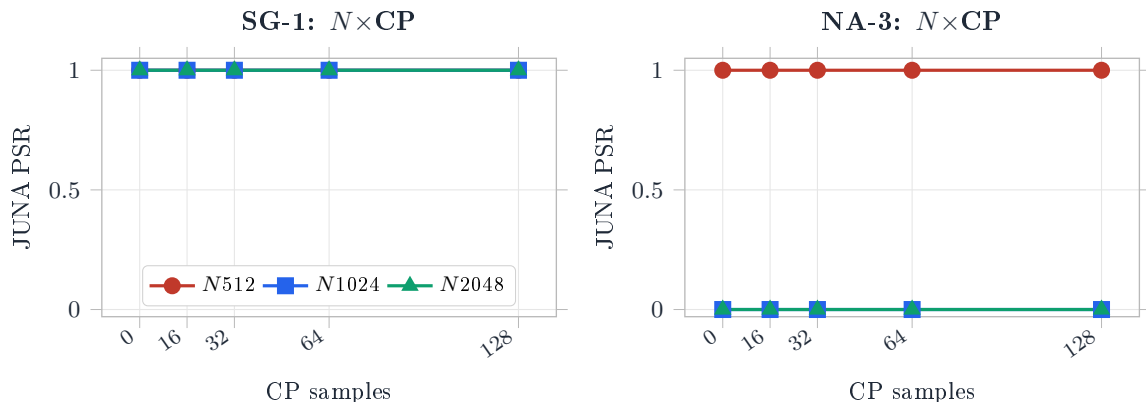


Figure 5: $N \times CP$ interaction. The best N is stable across CP within a channel but *differs between channels* (1024 on SG-1, 512 on NA-3), where larger N is essentially undecodable.

7 What This Means in Practice

The practical reading is that JUNA is the strongest receiver branch on most of the measured replay channels, while Partial FFT+FEC is usually the middle baseline but is not uniformly better than OFDM+FEC; on benign captures it can be worse when the extra adaptive degrees of freedom do not buy enough channel tracking improvement. For JUNA, N is the largest first determinant because it sets the OFDM time–frequency geometry and decides whether the packet can leave the decoding floor at all. Once N is working, the next limiting knob depends on the channel: some captures become pilot-limited, some FEC-limited, and some modulation-limited, plausibly reflecting whether the replay channel is more static or more time-varying. CP is usually a weaker knob in these sweeps. This is consistent with the role of JUNA: the decoder-aided update attacks the residual ICI and the associated off-diagonal channel coupling, so after the CP is adequate it is less often the dominant control variable.

8 Conclusion

Across twelve measured UWA-OFDM replay channels, the main result is that JUNA most often gives the highest packet-success rate among the three tested receiver branches. The gain is not merely a parameter-search artifact: when the same geometry is used, JUNA benefits from decoder-aided channel refinement that reduces residual ICI and off-diagonal coupling left by simpler one-tap or pilot-only updates. The strongest tuning lesson is also simple. Search the FFT size first, because N is the dominant determinant of whether JUNA reaches a usable packet-success regime. After that, tune the protection knobs—pilot density, FEC rate, and modulation—according to the channel. Partial FFT+FEC is a useful and often stronger baseline than plain OFDM+FEC, but it is not a guaranteed ordering on easy channels. The practical default is therefore not a single universal modem setting; it is a short search procedure: start from the transferable JUNA family, select N , and then choose the least aggressive protection settings that preserve packet delivery.